

NARRATIVE REVIEW OPEN ACCESS

Exposure to Airborne Particulate Matter in Regional Environments of Bangladesh and Its Effects on Respiratory Health: A Narrative Review

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Received: 14 February 2026 | **Revised:** 15 July 2026 | **Accepted:** 23 July 2026

Keywords: airborne particulate | Bangladesh | particulate matter | regional environments | respiratory health

ABSTRACT

Background: Air pollution represents a critical public health crisis in Bangladesh, driven by rapid industrialization, urbanization, and widespread emissions from brick kilns, traffic, and biomass burning. Elevated levels of fine particulate matter and toxic gases across both urban and rural regions contributes to significant reductions in life expectancy and escalating respiratory disease burdens. Therefore, the present review aims to examine the role of fine/ultrafine particulate matter and hazardous gases in respiratory health outcomes among the Bangladeshi population to synthesize evidence that can inform public health policy.

Methods: We included 42 studies to obtain data related to the fine/ultrafine particulate matter and hazardous gases and their impact on respiratory health in Bangladesh from scientific databases including PubMed, Google Scholar, Scopus, Directory of Open Access Journals, Web of Science, EMBASE, and the Cochrane Library between 2013 and 2025 (last 12 years).

Results: Findings indicated that PM_{2.5} concentrations in major cities such as Dhaka, Chattogram, and Khulna exceed WHO guidelines by 8 to 18 times, with consistent hotspots linked to industrial zones, brick kilns, transportation, and seasonal biomass burning. These pollutants are strongly associated with increased prevalence of asthma, chronic obstructive pulmonary disease, bronchitis, acute respiratory infections, and hospital admissions, particularly among children, the elderly, and occupational groups. The discussion underscores severe monitoring gaps in peri-urban and rural areas, where exposure remains under-characterized despite comparable pollution levels.

Conclusion: The findings of the present review indicates that the urgent regulatory enforcement, enhanced air quality monitoring, and targeted public health interventions are essential to mitigate respiratory health risks and reduce the nationwide burden of air pollution in Bangladesh. Furthermore, comprehensive geological and environmental monitoring studies are required across all districts to fully characterize exposure and guide effective public health interventions.

Abbreviations: BNAAQS, Bangladesh National Ambient Air Quality Standards; Cd, cadmium; CMAQ, community multiscale air quality model; CO, carbon monoxide; Cr, chromium; NO₂, nitrogen dioxide; O₃, ozone; PM, particulate matter; PM₁₀, particulate matter with aerodynamic diameter ≤ 10 micrometers (μm), includes both coarse and fine particles; PM₁₀, particulate matter with aerodynamic diameter ≤ 1.0 micrometer (μm); PM_{2.5}, particulate matter with aerodynamic diameter ≤ 2.5 micrometers (μm), also called fine particulate matter; Sc, scandium; SO₂, sulfur dioxide; UFP, ultrafine particles (diameter < 100 nm); V, vanadium; WHO, World Health Organization.

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1 | Introduction

Air pollution has emerged as one of the most pressing environmental and public health challenges of the 21st century in Bangladesh. Rapid industrialization, urban sprawl, uncontrolled vehicular growth, brick kiln expansion, and reliance on fossil fuel combustion have all contributed to deteriorating air population [1]. Bangladesh consistently places one of the top positions among the countries with the worst air quality globally [2]. Dhaka is also listed as one of the most polluted cities in the world, reflecting both local emissions and transboundary contributions [3]. However, air pollution in Bangladesh is not solely confined to the capital city Dhaka. Industrial hubs like Chattogram, Khulna, and Sylhet also report PM_{2.5} levels exceeding national standards [4].

Brick kiln workers, traffic police, rickshaw pullers, industrial laborers, and street vendors are continuously exposed to elevated pollutant concentrations, often without adequate respiratory protection [5, 6]. Similarly, garment industry workers in poorly ventilated factories, as well as agricultural laborers exposed to pesticide drift and biomass smoke, face compounding environmental health risks. Children, the elderly, and those living in slum settlements close to industrial zones face higher pollutant burdens due to both exposure duration and physiological vulnerability [7].

In addition, there is a growing body of evidence linking particulate exposure to lung cancer incidence and mortality in Bangladesh [8]. Particularly vulnerable populations include children under five, older adults and individuals with preexisting cardiovascular disease [4]. Studies have also shown that urban air pollution accounts for roughly 10% of respiratory illnesses in Bangladesh, underscoring the direct health service burden [9]. Moreover, the deployment of the Clean Air Act (CAA) and related regulatory frameworks remains nascent, and enforcement suffers from institutional capacity limitations, resource constraints, and political economy complexities [10].

Equally concerning is the limited coverage of air quality monitoring systems. In sites like Netrokona district, ambient PM concentrations rival national hotspot levels, yet lack representation in national monitoring networks [11]. On the epidemiological front, although previous study linked perceived air pollution to respiratory symptoms and mental distress among urban residents, there remained a shortage of longitudinal cohort studies that could robustly capture chronic exposure effects across health, cognitive, and socioeconomic dimensions [12].

Together, these research priorities represent critical pathways for creating an evidence-informed framework that strengthens regulation, improves environmental governance, and safeguards public health in Bangladesh. Therefore, the review aimed to explore the role of ultrafine particulate and toxic gases in the respiratory health problems in the Bangladeshi population. The findings of the study will provide the insights to develop the public health policy in Bangladesh.

2 | Materials and Methods

2.1 | Data Searching

Data related to the impacts of urban air pollution on respiratory health in Bangladesh, with a focus on ultrafine particulate matter and toxic gaseous emissions, were retrieved from multiple online scientific databases. PubMed, Google Scholar, Scopus, Directory of Open Access Journals (DOAJ), Web of Science, EMBASE, and the Cochrane Library were searched to collect data between 2013 and 2025 (last 12 years). Only peer-reviewed journal articles in English that met predetermined inclusion criteria were considered. The search strategy incorporated combinations of keywords and Boolean operators, including: “air pollution OR urban air pollution”, “ultrafine particulate matter OR PM_{0.1}”, “particulate matters OR PM_{2.5} OR PM₁₀”, “toxic gases OR NO_x OR Sox OR CO OR O₃”, “respiratory diseases AND lung cancer”, “asthma AND COPD”, “Bangladesh” etc [13]. Reference lists of eligible articles were also screened to identify additional studies.

2.2 | Eligibility Criteria

This review included articles based on predetermined inclusion and exclusion criteria. This study included articles if they; (a) focused on urban air pollution in Bangladesh, (b) examined health outcomes related to ultrafine particulates and toxic gases, and (c) provided epidemiological or experimental evidence linking pollution exposure to respiratory health impacts. However, articles were excluded if they; (a) relied solely on animal models, (b) were commentaries, editorials, conference abstracts, or articles without robust data, (c) lacked quantitative or qualitative evidence on the biological, social, or economic dimensions of air pollution, or (d) were not peer-reviewed.

2.3 | Study Selection Process and Data Extraction

We retrieved 202 articles from several online databases. After the elimination of duplicate articles, 126 studies were short-listed for examining abstracts and titles. However, 49 of them were not related to the aim of our study. Therefore, 77 studies were selected for full-text review. Commentaries, letters to editors, and studies with a lack of sufficient data were excluded after reviewing the full text of all selected studies. Studies with low quality were also excluded from the current study. Finally, data were extracted from 42 research articles based on study location, pollutant types, health outcomes, socio-economic effects, climate implications, and proposed mitigation measures. Two separate authors independently retrieved and extracted data based on eligibility criteria. Any anomalies found during the data collection were solved through detailed discussion between them. For the preliminary selection of the included articles, basic details of the papers were noted, including the name of the author, study title, year of publication, abstract, objectives, findings and indexing of journals. Furthermore, we prepared a checklist based on Newcastle-Ottawa Scale containing the fundamental characteristics of the included studies to ensure the quality and reliability of the included studies. The checklist included a clear statement regarding the representativeness of the study area and population, methodological

transparency and findings assessment method (Supporting Information S1: Table S1). We modified the NOS scale as no NOS version exists specifically for environmental/air quality exposure studies. However, it retains the original 3 domains such as selection, comparability, and outcome/exposure domains and 9-point scoring structure that reflects constructs relevant to spatial/population representativeness, sampling adequacy, control for meteorological and land-use confounders, and the validity of pollutant measurements. Studies that did not meet the expected data collection requirements, they were excluded from the current study. We assessed the quality of each study by scoring 1 for “Yes” and 0 for “No” or “Unclear.” We categorized studies reporting our intended research outcome as low, moderate, or high quality on the basis of the obtained score. Studies scoring 0–4, 5–7, and 8–9 were classified as low-, moderate-, or high-quality studies, respectively. We only included studies that obtained moderate to high grade of quality.

3 | Results and Discussions

3.1 | Particulate Matters and Toxic Gas Concentrations in All the Major Cities in Bangladesh

Dhaka city showed 18 times greater $PM_{2.5}$ concentration than the WHO annual guideline, revealed a steady upward trend between 2003 and 2019 [14]. Meteorological studies explained that nearly 57% of daily $PM_{2.5}$ level variation was found in Dhaka, where temperature and humidity were the prime indicators [15]. Besides, the TROPOMI satellite and NO_2 maps identified persistent urban NO_2 hotspots across the Dhaka metropolitan area [16]. Dhaka and nearby Gazipur consistently formed a contiguous $PM_{2.5}$ along with NO_2 hotspot, with seasonal winter peaks that exacerbated city-wide exposure burdens [17]. Brick kilns, biomass burning, and traffic were identified as dominant contributors of this high concentration of particulate matters and toxic gas in Dhaka city [13]. Green spaces and reduced traffic mitigate pollution, but decreased vegetation coverage correlates with rising $PM_{2.5}$ [18]. For instance, full lockdowns during COVID-19 reduced the $PM_{2.5}$ level by approximately 29% in Dhaka, where reduced traffic or urban activity were considered as major drivers [19]. In the winter-time, the black carbon (BC) based inhaled dose were surged in Dhaka city, which amplified the short-term cardiopulmonary risk [20].

Moreover, a satellite-based NO_2 mapping combined with local monitors highlighted persistent NO_2 hotspots in Dhaka, Chattogram, Sylhet and Rajshahi, which were further linked with vehicular and industrial emissions across those urban centers [21]. Wintertime $PM_{2.5}$ levels in Chattogram often exceed $150 \mu g/m^3$ which suggested strong seasonal and diurnal variation [22]. Moreover, amplified PM_{10} and BC fractions in wintertime at Chattogram indicated strong seasonal emission and meteorological controls [23]. In Khulna city, a monitoring of particulate matter concentrations in coastal regions revealed maximum $PM_{1.0}$, $PM_{2.5}$ and PM_{10} level that demonstrated elevated health risks for Bangladeshi population in terms of habitancy [24]. In Barisal, the average $PM_{2.5}$ concentration was approximately eight times higher than the World Health Organization's

recommended daily average [25]. Mymensingh suffered nearly same magnitudes of $PM_{2.5}$ exposure as large cities [26]. Along with the Mymensingh-Gazipur Highway in winter, $PM_{2.5}$ level primarily consisting of ultrafine particles [27].

The secondary cities such as Rajshahi and Rangpur showed a rising particulate matter trends which were tied to urbanization and local industry [28]. Besides, Dinajpur district town assessed the concentration of particulate matter in relation to various land uses, especially at elevated levels in industrial areas [29]. Narsingdi city also experienced a significant air pollution, with average $PM_{2.5}$ concentrations exceeding WHO guidelines [30]. A long-term satellite-based re-analysis showed higher aerosol optical depth in winter or spring, which were produced through anthropogenic aerosols, was dominated by urban load [31]. Trace-metal studies of particulate matter reported high enrichment factors, indicating that industrial and combustion sources were the main contributors [32].

The CMAQ and local models reproduced the background of $PM_{2.5}$ level in Bangladesh, but it often underpredicted extreme pollution episodes [33]. The ultrafine particles (UFP; particle size less than 100 nanometer) showed a large intra-urban particle number concentration gradient that indicated the much higher traffic sites residing into the cities of Bangladesh [34]. The $PM_{2.5}$ level-bound trace elements such as S, V, Cr, Cd etc. deposited in lungs highlighted potential impacts on respiratory health based on the quantified deposited dose [35]. In broader sense, particulate matter such as $PM_{1.0}$, $PM_{2.5}$, and PM_{10} , as well as toxic gases such as NO_2 , SO_2 , CO, O_3 in major Bangladeshi cities exceeded in both WHO and BNAQs claimed accepted range [36].

3.2 | Causes of Air Pollution in Major Cities of Bangladesh

The Dhaka city exhibited a substantial annual elevation in $PM_{2.5}$ concentrations primarily attributed to traffic emissions and industrially generated ultrafine particulates. Ultrafine particles emitted from extensive industrial zones are responsible for deteriorating air quality in Gazipur. The $PM_{2.5}$ concentrations observed in the city to be approximately two to three times higher than urban background levels [37]. In Narayanganj, emissions from textile and tannery operations, along with shipping-related activities, add chemicals and trace metals to the air. Pollution levels show distinct daytime peaks that line up with worse air quality index readings at several monitoring locations [35].

During the dry season, port and shipping emissions in major cities such as Chittagong are responsible for seasonal spikes in air pollution, driven by elevated ultrafine particle concentrations [38]. Similarly, refinery and shipbreaking activities in Khulna contributes to increased ultrafine particle levels, largely due to metal enrichment, thereby exacerbating local air quality deterioration [24]. In Barisal, episodic increases in air pollution are observed, primarily driven by emissions from inland water transport and biomass combustion [25]. In Rangpur, a study identified that industrial and residential areas have higher concentrations of particulate matter, indicating significant

pollution levels in these zones [39]. Brick kiln operations and construction-related dust significantly degrades air quality in Rajshahi, with PM_{2.5} levels at certain sites surpassing permissible thresholds [28]. High traffic density, unpaved roads, and the use of outdated vehicles contributes to elevated levels of air pollutants in Mymensingh [40].

Air pollution in Dinajpur was amplified by small industrial activities and the re-suspension of waste, resulting in concentrations above typical rural baseline levels [29]. In Sherpur, peak activity during local market hours is associated with elevated ultrafine particulate emissions from local transport, resulting in heightened air pollution levels [41]. Wintertime smog in Sirajganj is significantly impacted by emissions from combined industrial and brick-kiln areas, leading to elevated levels of particulate pollution [42].

In the western region of Bangladesh, small-scale industrial hotspots in Meherpur were identified as major contributors to air pollution, with particulate levels approximately twice as high as background concentrations between 2019 and 2022 [43]. In Chuadanga, seasonal biomass burning and combined freight traffic showed episodic air contamination due to ultrafine particles during harvesting months [44]. Industrial zones in Jessore exhibits two and half times higher particulate matter levels relative to rural areas [45]. In Narail, traffic congestion during peak hours is the primary driver of ultrafine particle increases, elevating concentrations by 4 to 6 times [46]. Post-harvest biomass burning in Pabna district further contributes to air pollution, raising particulate matter levels by 30%–40% at several monitoring locations [47]. In Kushtia, nocturnal ultrafine particle emissions resulting from combined traffic and small-scale industrial activities during the early morning hours are identified as a major source of air pollution [48].

In eastern Bangladesh, traffic emissions in Brahmanbaria elevates ultrafine particle concentrations, leading to moderate to severe air pollution [49]. Factory corridors in Feni recorded 1.6–1.9 times higher fine particle levels, resulting in frequent smog events [50]. In Munshiganj, construction-related emissions caused BNAAQS exceedances in work zones, producing polluted air plumes over nearby regions [51]. Sea breeze dynamics played a crucial role in modulating ultrafine particle concentrations along the coastal corridors of Lakshmipur [52]. The onshore breeze likely facilitated the inland transport and dispersion of marine aerosols, synchronizing with local emissions to shape distinct diurnal patterns of particle accumulation. In contrast, episodic pollution events in Sunamganj were primarily associated with wetland desiccation and intensified local combustion activities during the winter season [53], reflecting a strong coupling between hydrometeorological variability and emission-driven processes. Furthermore, Narsingdi exhibited persistent air quality degradation characterized by concurrent elevations in NO₂ and ultrafine particle concentrations, attributed to emissions from textile processing units and dense industrial corridors [17].

Globally, air pollution remains a pervasive and largely unresolved public health and environmental challenge. The annual PM_{2.5} benchmark set by the WHO's air quality guideline is 5 µg/m³ [14]. However, the current study recorded the

greater concentration of this threshold in the case of all major cities in Bangladesh. Similarly, more than 90% of cities from 138 countries such as Chad, Pakistan, DR Congo, and India crossed the threshold value set by WHO. Only 17% of cities worldwide met the WHO's annual PM_{2.5} guideline [54, 55]. Central and South Asia hosted the majority of the world's most polluted metropolitan areas [55]. Therefore, a sustained and centrally enforced policy action on industrial emission controls, vehicle retirement schemes, and clean-energy adoption is required to reduce the high concentration of particulate matter and toxic gases. In China, this type of action reduced the urban pollution in Beijing's annual PM_{2.5} concentration from roughly 90 µg/m³ in 2013 to below 30 µg/m³ by 2024 [56].

3.3 | Air Pollution and Respiratory Health Issues in Major Cities of Bangladesh

Air pollution in cities of Bangladesh is critically high, primarily caused by fine particulate matter and toxic gases. These pollutants are linked to significant health issues, especially respiratory diseases such as asthma, bronchitis, and chronic obstructive pulmonary disease (COPD). In the northern region, respiratory diseases, such as asthma, were notably prevalent among children, while chronic respiratory diseases like COPD were common in adults living in small industrial zones or areas with high vehicular emissions [57]. Hospital admission data from these areas showed a rise in respiratory-related illnesses, especially among vulnerable populations such as children and the elderly [58]. The western regions of Bangladesh, including Rajshahi and Khulna, suffered from the cumulative effects of industrial emissions, including brick kilns and refineries, which contributed significantly to particulate pollution [24]. These cities, along with Rajshahi, showed an upward trend in respiratory disease cases, with hospital data revealing a surge in COPD and bronchitis among adults [28]. In the southern part of Bangladesh, cities like Barisal and Khulna faced air pollution primarily from river ports, industrial activities, and seasonal biomass burning. This pollution was closely linked to an increase in respiratory infections among the local population, as indicated by hospital admission rates. The coastal areas of the southern region also experienced poor air quality due to the frequent burning of agricultural waste and the concentration of emissions from inland water transport [53].

Finally, in the eastern region, cities such as Brahmanbaria and Feni reported high levels of air pollution, particularly from industrial corridors and vehicular emissions. Airborne particulate matter in these cities significantly impacted the respiratory health of both children and adults [49]. Similarly, the global health burden associated with air pollution is substantial. For instance, particulate matter from fossil-fuel combustion alone contribute to the cardiovascular, respiratory, and lung-cancer-related mortality globally [59]. Table 1 represents the data regarding the comparative levels of particulate matters and respiratory health in major cities of Bangladesh. According to the Bangladeshi ambient air quality standards, the permissible concentration ranges are 15–65 µg/m³ for PM_{2.5} and 50–150 µg/m³ for PM₁₀ in airborne particulate matter [47].

TABLE 1 | Comparative levels of particulate matters and respiratory health in major cities of Bangladesh.

City	Region	PM _{2.5} ($\mu\text{g}/\text{m}^3$)	PM ₁₀ ($\mu\text{g}/\text{m}^3$)	Respiratory diseases	Hospital admissions	References
Dhaka	Central	80–100	120–150	Asthma, COPD, Bronchitis	High	[12]
Mymensingh	Central	90–110	150–180	Asthma, Bronchitis	High	[26]
Gazipur	Central	100–140	100–170	Respiratory Infections	High	[17]
Narsingdi	Central	30–80	50–90	Asthma, Bronchitis	High	[17]
Dinajpur	North	70–90	90–120	Coughing, Asthma	Moderate	[29]
Pabna	North	20–60	50–100	Respiratory Infections	Moderate	[47]
Brahmanbaria	East	10–70	20–220	Asthma, Coughing	Moderate	[49]
Feni	East	20–30	60–70	Asthma, Bronchitis	Moderate	[48]
Lakshmipur	East	100–120	100–210	Coughing, Asthma	Moderate	[52]
Khulna	West	350–400	300–350	Acute Respiratory Infections	High	[24]
Rajshahi	West	180–220	250–300	Coughing, Shortness of Breath	Moderate	[28]
Kushtia	West	60–90	129–159	Asthma, Bronchitis	Moderate	[48]
Jessore	West	20–50	30–85	Asthma, Bronchitis	Moderate	[45]
Chattogram	South	100–150	80–120	Coughing, Shortness of Breath	Moderate	[22]
Narail	South	60–70	80–100	Asthma, Bronchitis	Moderate	[46]

Note: PM_{2.5} and PM₁₀ ranges represent the minimum and maximum mean concentrations reported in the cited source for each city over the study period covered by that source (see individual references for exact years and averaging basis); Respiratory Diseases lists the disease types reported as most prevalent in the cited source, and “Hospital Admissions” (High/Moderate) reflects the relative admission-load categorization used by the cited source. Where pollution and health data in a row derive from different studies within the same source document, both are covered by the single citation given.

3.4 | Limitations of the Study

A few drawbacks of this study should be acknowledged when interpreting the findings. Firstly, the findings may not be broadly applicable due to the limited studies available representing the data regarding region specific, particulate matter exposure, air quality and respiratory health in Bangladesh. Secondly, as it is a narrative review, it lacks the quantitative synthesis of data associated with systematic review that may affect the comparability of findings. Thirdly, variations in seasonal and geographical pollution patterns may not reflect the whole scenario of the impact of particulate matters exposure and toxic gases in respiratory health in Bangladesh. Finally, there is a profound lack of longitudinal, cohort-based epidemiological studies that can track chronic health outcomes, cognitive impacts, and socioeconomic consequences over time, especially among high-risk occupational groups like brick kiln workers, traffic police, and garment laborers.

4 | Conclusion

This review underscores the critical and escalating public health emergency posed by air pollution across Bangladesh that directly linked to widespread exposure of particulate matters and toxic gases. The findings indicated that higher particulate matter exposure in major cities adversely affects respiratory health in Bangladesh. Findings indicated that PM_{2.5} concentrations in Dhaka, Chattogram, and Khulna exceed WHO guidelines by 8 to 18 times. This higher concentration of PM and toxic gases was strongly associated with increased

prevalence of asthma, chronic obstructive pulmonary disease (COPD), bronchitis, acute respiratory infections, and hospital admissions, particularly among children, the elderly, and occupational groups. Industrial zones, brick kilns, transportation, and seasonal biomass burning were identified as hotspots linked to particulate matters exposure. The findings of the study underscore severe monitoring gaps in peri-urban and rural areas, where exposure remains under-characterized despite comparable pollution levels. Therefore, the urgent regulatory enforcement, enhanced air quality monitoring, and targeted public health interventions are essential to mitigate respiratory health risks and reduce the nationwide burden of air pollution in Bangladesh. Furthermore, comprehensive geological and environmental monitoring studies are required across all districts to fully characterize exposure and guide effective public health interventions. Epidemiologically, large-scale, long-term cohort studies should be initiated to establish definitive causal links between specific pollution profiles such as PM_{2.5}, NO₂, and chronic cardiorespiratory diseases, developmental issues in children, and economic productivity loss.

Author Contributions

K. M. Ferdousul Haque: conceptualization, data curation, investigation, methodology, writing – original draft. **Mohammad A Rashid:** formal analysis, investigation, methodology, validation, project administration, writing – review and editing. **Mohammad Sharifur Rahman:** investigation, formal analysis, project administration, visualization, writing – review and editing, validation. **Md. Khalid Hossain:** methodology, formal analysis, validation, resources,

visualization, writing – review and editing. **Md. Shafiul Hossen:** conceptualization, data curation, investigation, methodology, project administration, supervision, writing – original draft.

Acknowledgments

AI has been used for language editing and grammar check.

Funding

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

Transparency Statement

The corresponding author certifies that no significant details have been left out and that this paper is an honest, accurate, and transparent account of the work being described. All authors have read and approved the final version of the manuscript.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: hsr272925-sup-0001-Supplementary_Table_1_most_recently_revised_clean_version.docx.